

Response Letter with Annotations

Clauset, A., Arbesman, S., & Larremore, D. B. (2015). Systematic inequality and hierarchy in faculty hiring networks. Science Advances, 1(1), e1400005.
<https://doi.org/10.1126/sciadv.1400005>

Annotations

1. Purposes

a. Motivation (one-sentence summary)

They want to study the faculty hiring system about which university hires which university's graduates.

b. Specific Rationale (several bullet points)

- Previous literature doesn't systematic study the common patterns and efficiencies of faculty hiring across disciplines.
- Hiring decisions incorporate collective judgments of institutional quality
- Prestige can influence academic outcomes, such as hiring process, but it is difficult to measure it objectively.
- Existing college ranking has some limitations.

c. Research Questions and/or Hypotheses

• Identify IV, DV, mediator, moderator

1. IV: institutional prestige, gender
2. DV: placement, faculty production, faculty placement, institutional influence, social inequality
3. Moderator: NA
4. Mediator: NA

• Identify specific relationships among variables

They didn't explicitly propose research questions or hypotheses. I think researchers aim to explore patterns inside the hiring network.

2. Method

a. Research Design

- Network analysis based on cross-sectional observational data

b. Sample (representative sample?)

- Nearly 19000 tenure-track or tenured faculty in US.
- I think it is close to a representative sample since they select three different disciplines, but it limits to US.

c. Measures

- Prestige is measured by minimum violation ranking
- Institutional influence is measured by network centrality
- Predicatory accuracy is measured by area under the curve (AUC)
- Social inequality is measured by Gini coefficient
- Rest of variables are directly available in the data set

d. Analysis

- They build hiring networks for a network analysis

1. Nodes are institutions, edges are faculty hire (from = doctoral institution, to = hiring institution)

3. Results and Conclusions

a. Answers to RQs and Hypotheses

- Increased institutional prestige leads to increased faculty production, better faculty placement, and a more influential position within the discipline.
- Faculty hiring system has a hierarchy in hiring faculty, which means most of faculty place at institutions less prestigious than where they earn their PhD.
- There is an inequality in faculty production, with only 25% of institutions producing 71 to 86% of all tenure-track faculty.
- In business and computer science, elite women place worse than elite men.

b. Include Key Figures/Tables

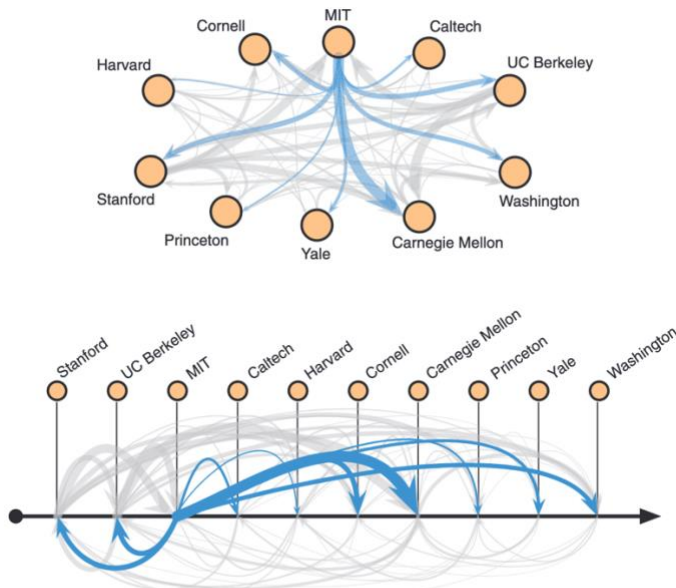


Fig. 1. Prestige hierarchies in faculty hiring networks. (Top) Placements for 267 computer science faculty among 10 universities, with placements from one particular university highlighted. Each arc (u,v) has a width proportional to the number of current faculty at university v who received their doctorate at university u ($u \neq v$). **(Bottom)** Prestige hierarchy on these institutions that minimizes the total weight of "upward" arcs, that is, arcs where v is more highly ranked than u .

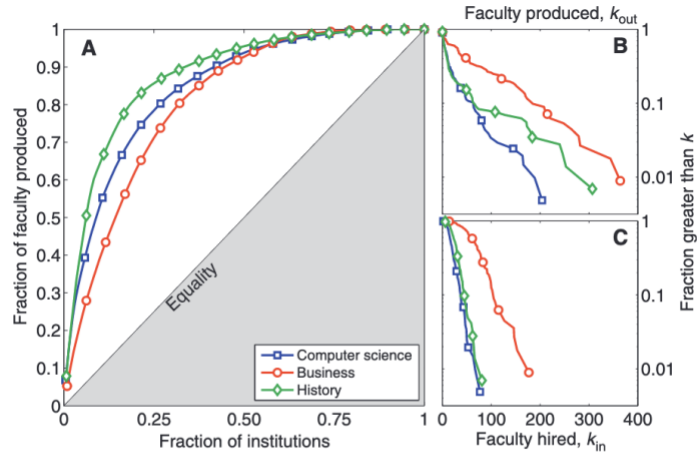


Fig. 2. Inequality in faculty production. (A) Lorenz curves showing the fraction of all faculty produced as a function of producing institutions. (B and C) Complementary cumulative distributions for institution out-degree (faculty produced) and in-degree (faculty hired). The means of these distributions are 21 for computer science, 70 for business, and 29 for history.

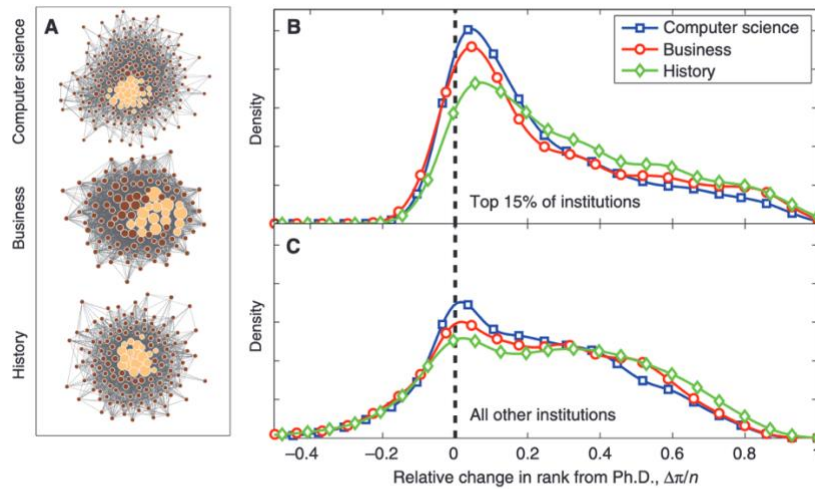


Fig. 3. Faculty placement distributions. (A) Network visualizations for computer science, business, and history (top to bottom) showing central positions for institutions in the top 15% of prestige ranks (highlighted; vertex size proportional to k_o). (B and C) Estimated probability density functions for relative change in prestige (doctoral to faculty institution) for (B) the top 15% and (C) the remaining institutions, showing a common but right-skewed structure.

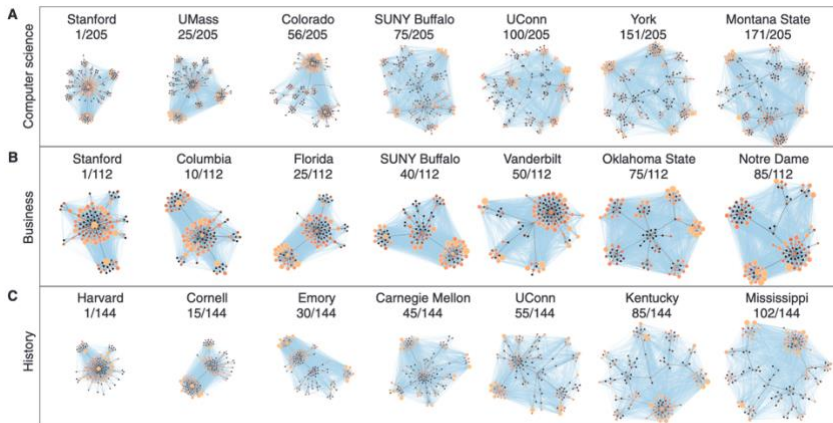


Fig. 4. Core-periphery patterns. (A to C) For several institutions within each disciplinary hiring network, we highlight the tree of shortest paths rooted at each u within this network (black) for (A) computer science, (B) business, and (C) history (vertex size is proportional to out-degree, and lighter colors indicate higher prestige). As prestige increases (left), the paths in these trees contract, reflecting a more central network position, increased faculty production, and better faculty placement.

c. Authors' Conclusions

Prestige is a powerful factor in influencing faculty hiring. Hiring system has a stale hierarchy, and faculty production is not equal.

4. Discussion

a. Implications

- This network-based prestige may be a better than other existing rankings.
- Possible application to measure program quality or other areas.

b. Limitations

- They only focus on cross-sectional data at one point in time.
- Limited to only three disciplines and to US
- Other factors may overlap with prestige, such as merit and status. Current operationalization of prestige can't exclude other factors such as geography.
- Missing individual-level factors
- No causality

c. Future Directions

- Identify why hierarchy is formed.
- Examine other factors in doctoral program, such as mentoring, publication, hiring situations.
- Apply experiment to test causality.

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Response Letter (~600 words)

This article systematically examines faculty hiring in the US based on three different disciplines. Across computer science, business, and history, faculty hiring follows a strong, hierarchical pattern. Only 9–14% of faculty are hired by institutions more prestigious than where they earned their PhD. This indicates a system where downward mobility is the norm. The results are not very unexpected and seem to validate this common knowledge.

As authors discuss, this article has many strengths. Their findings have many theoretical contributions to social hierarchy in academia. Moreover, the study uses a large number of faculty across many PhD-granting institutions, producing one of the most complete hiring datasets ever assembled. This gives the findings strong empirical grounding. In addition, their operationalizations are very clear and transparent given the rationales. They also have rigorous statistical testing for the validity of claims about hierarchy, inequality, and predictive power.

Other than what they indicate in the article, I think the most innovative point is their method, which follows a relatively novel approach of network analysis. Their original data is very simple and only with several columns. The difficult part is to create operationalization of each variable. Authors apply different network features to their variables. For instance, network centrality is used to measure institutional. This operationalization is consistent with other readings such as Zhao et al. (2022) this week to measure influence. Edge is used to measure hiring decision. Additionally, some measures are based on network features developed by authors, such as ranking change. Other measures are borrowed based on existing literature, such as minimum violation ranking for prestige, and out-degree for inequality. I especially like this part that they apply basic network features in this specific research context.

Besides strengths, the research also has some weaknesses. I think the biggest concern is their operationalization on prestige. Currently the measure blends with other potential confounders, such as geographic advantages, social status. Additionally, as they found some exceptional cases, it will be interesting to focus on these cases and explore why these happen. Potentially they are due to individual factors. Some weaknesses are because not reflecting real life situations. For example, they leave out individual factors such as mentoring, postdocs, and hiring situations. And the data only include only PhD-granting universities but no other institutions such as liberal arts colleges and teaching schools. The study also treats each discipline as a closed system, even though real hiring often crosses fields. Besides gender, I think other factors could also be explored in the future, such as nationality and race, since there are so many international scholars now in academia.

Despite this weakness, I think this article is very inspiring and still holds a really positive opinion about this article overall, because authors already did their best on their research focus (not because it is published on Science). Connected with other readings, I think other topics are also available for network analysis, even if “from” and “to” are not individual, for example, to study information diffusion (Gong et al., 2023).